Experiment No.5

Title : Program to implement package and sub packages in java

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1. Library Management System

Problem Statement: Create a package library that contains classes like Book, Member, and LibraryManagement to handle book lending, returns, and member management.

Book.java

File Edit View

```
package library;

public class Book {
    int bookId;
    String title;
    String author;
    int isLent;  // 1 for lent, 0 for available

    public Book(int bookId, String title, String author) {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
        this.isLent = 0; // Initially available
    }

    public void lendBook() {
        if (isLent == 0) {
            isLent = 1;
            System.out.println(title + " has been lent.");
        } else {
            System.out.println(title + " is already lent.");
        }
    }

    public void returnBook() {
        if (isLent == 1) {
            isLent = 0;
```

Ln 1, Col 1 1,002 characters 150% Windows (CRLF) UTF-8

```
    public void returnBook() {
        if (isLent == 1) {
            isLent = 0;
            System.out.println(title + " has been returned.");
        } else {
            System.out.println(title + " was not lent.");
        }
    }

    public void displayBook() {
        System.out.println("Book ID: " + bookId + ", Title: " + title + ", Author: " + author + ", Status: " + (isLent == 1 ?
"Lent" : "Available"));
    }
}
```

Ln 1, Col 1 1,002 characters 150% Windows (CRLF) UTF-8

Book.java Member.java +

File Edit View

```java
package library;

public class Member {
    int memberId;
    String name;

    public Member(int memberId, String name) {
        this.memberId = memberId;
        this.name = name;
    }

    public void displayMember() {
        System.out.println("Member ID: " + memberId + ", Name: " + name);
    }
}
```

Book.java Member.java LibraryManagement.java +

File Edit View

```java
package library;

public class LibraryManagement {
    public void issueBook(Book book) {
        book.lendBook();
    }

    public void returnBook(Book book) {
        book.returnBook();
    }
}
```

Book.java Member.java LibraryManagement.java LibraryMain.java +

File Edit View

```java
import library.Book;
import library.Member;
import library.LibraryManagement;

public class LibraryMain {
    public static void main(String[] args) {
        Book book1 = new Book(101, "Java Programming", "James Gosling");
        Member member1 = new Member(1, "Alice");

        LibraryManagement lib = new LibraryManagement();

        System.out.println("\n--- Library System ---");
        book1.displayBook();
        member1.displayMember();

        System.out.println("\nIssuing Book...");
        lib.issueBook(book1);
        book1.displayBook();

        System.out.println("\nReturning Book...");
        lib.returnBook(book1);
        book1.displayBook();
    }
}
```

Ln 1, Col 1 | 678 characters | 160% | Windows (CRLF) | UTF-8

```
Microsoft Windows [Version 10.0.26100.3624]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Shreya>cd C:\Java Programs\Library

C:\Java programs\Library>javac -d . Book.java

C:\Java programs\Library>javac -d . Member.java

C:\Java programs\Library>javac -d . LibraryManagement.java

C:\Java programs\Library>javac LibraryMain
error: Class names, 'LibraryMain', are only accepted if annotation processing is explicitly requested
1 error

C:\Java programs\Library>javac LibraryMain.java

C:\Java programs\Library>java LibraryMain

--- Library System ---
Book ID: 101, Title: Java Programming, Author: James Gosling, Status: Available
Member ID: 1, Name: Alice

Issuing Book...
Java Programming has been lent.
Book ID: 101, Title: Java Programming, Author: James Gosling, Status: Lent

Returning Book...
Java Programming has been returned.
Book ID: 101, Title: Java Programming, Author: James Gosling, Status: Available

C:\Java programs\Library>
```

2. Employee Payroll System

Problem Statement: Develop a package payroll with classes such as Employee, Salary, and Payslip to calculate employee salaries, deductions, and generate pay slips.

```java
package payroll;

public class Employee {
    private int empId;
    private String name;
    private double basicSalary;

    public Employee(int empId, String name, double basicSalary) {
        this.empId = empId;
        this.name = name;
        this.basicSalary = basicSalary;
    }

    // Getter method to access basicSalary
    public double getBasicSalary() {
        return basicSalary;
    }

    public void displayEmployee() {
        System.out.println("Employee ID: " + empId + ", Name: " + name + ", Basic Salary: " + basicSalary);
    }
}
```

```java
package payroll;

public class Salary {
    double basicSalary;
    double hra;
    double da;
    double pf;
    double grossSalary;
    double netSalary;

    public Salary(double basicSalary) {
        this.basicSalary = basicSalary;
    }

    public void calculateSalary() {
        hra = 0.20 * basicSalary;  // HRA = 20% of basic salary
        da = 0.10 * basicSalary;   // DA = 10% of basic salary
        pf = 0.12 * basicSalary;   // PF = 12% of basic salary

        grossSalary = basicSalary + hra + da;
        netSalary = grossSalary - pf;  // Deducting PF
    }

    public void displaySalary() {
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("PF Deduction: " + pf);
        System.out.println("Gross Salary: " + grossSalary);
        System.out.println("Net Salary: " + netSalary);
    }
}
```

```java
package payroll;

public class Payslip {
    Employee employee;
    Salary salary;

    public Payslip(Employee employee, Salary salary) {
        this.employee = employee;
        this.salary = salary;
    }

    public void generatePayslip() {
        System.out.println("\n--- PAYSLIP ---");
        employee.displayEmployee();
        salary.displaySalary();
    }
}
```

```java
import payroll.Employee;
import payroll.Salary;
import payroll.Payslip;

public class PayrollMain {
    public static void main(String[] args) {
        Employee emp1 = new Employee(101, "John Doe", 50000);
        Salary salary1 = new Salary(emp1.getBasicSalary()); // Use getter method

        salary1.calculateSalary();

        Payslip payslip1 = new Payslip(emp1, salary1);
        payslip1.generatePayslip();
    }
}
```

```
C:\Java programs>cd C:\Java Programs\Payroll

C:\Java programs\Payroll>javac -d . Employee.java

C:\Java programs\Payroll>javac -d . Salary.java

C:\Java programs\Payroll>javac -d . Payslip.java

C:\Java programs\Payroll>javac PayrollMain.java

C:\Java programs\Payroll>java PayrollMain

--- PAYSLIP ---
Employee ID: 101, Name: John Doe, Basic Salary: 50000.0
Basic Salary: 50000.0
HRA: 10000.0
DA: 5000.0
PF Deduction: 6000.0
Gross Salary: 65000.0
Net Salary: 59000.0

C:\Java programs\Payroll>
```

3. Banking System

Problem Statement: Create a package banking with classes like Account, Transaction, and Customer to handle deposits, withdrawals, and account management.

Account.java | Transaction.java | Customer.java | BankingSystem.java

File Edit View

```java
package banking;

public class Account {
    String accountNumber;
    double balance;

    public Account(String accountNumber, double balance) {
        this.accountNumber = accountNumber;
        this.balance = balance;
    }

    public void deposit(double amount) {
        balance += amount;
    }

    public void withdraw(double amount) {
        if (balance >= amount) {
            balance -= amount;
        } else {
            System.out.println("Insufficient funds!");
        }
    }
}
```

Account.java | Transaction.java | Customer.java | BankingSystem.java

File Edit View

```java
package banking;

public class Transaction {
    public static void record(String type, double amount) {
        System.out.println("Transaction: " + type + " of amount " + amount);
    }
}
```

Account.java | Transaction.java | Customer.java | BankingSystem.java

File Edit View

```java
package banking;

public class Customer {
    String name;
    Account account;

    public Customer(String name, String accountNumber, double balance) {
        this.name = name;
        this.account = new Account(accountNumber, balance);
    }

    public void deposit(double amount) {
        account.deposit(amount);
        Transaction.record("Deposit", amount);
    }

    public void withdraw(double amount) {
        account.withdraw(amount);
        Transaction.record("Withdrawal", amount);
    }

    public void showInfo() {
        System.out.println("Customer: " + name);
        System.out.println("Account Number: " + account.accountNumber);
        System.out.println("Balance: " + account.balance);
    }
}
```

Ln 16, Col 1 | 726 characters | 160% | Windows (CRLF) | UTF-8

```java
import banking.Account;
import banking.Transaction;
import banking.Customer;

public class BankingSystem {
    public static void main(String[] args) {
        Customer c = new Customer("Ram", "123456", 1000.0);

        c.showInfo();

        c.deposit(500);
        c.withdraw(300);

        c.showInfo();
    }
}
```

```
C:\Java programs\Banking>javac -d . Account.java

C:\Java programs\Banking>javac -d . Transaction.java

C:\Java programs\Banking>javac -d . Customer.java

C:\Java programs\Banking>javac BankingSystem.java

C:\Java programs\Banking>java BankingSystem
Customer: Ram
Account Number: 123456
Balance: 1000.0
Transaction: Deposit of amount 500.0
Transaction: Withdrawal of amount 300.0
Customer: Ram
Account Number: 123456
Balance: 1200.0

C:\Java programs\Banking>
```

4. Student Management System

Problem Statement: Implement a package student containing classes like Student, Marks, and ReportCard to manage student records, marks, and generate report cards.

```java
package student;

public class Student {
    String name;
    int rollNumber;

    public Student(String name, int rollNumber) {
        this.name = name;
        this.rollNumber = rollNumber;
    }

    public void showStudentInfo() {
        System.out.println("Student Name: " + name);
        System.out.println("Roll Number: " + rollNumber);
    }
}
```

Student.java | Marks.java | ReportCard.java | SchoolSystem.java

File Edit View

```java
package student;

public class Marks {
    int math, science, english;

    public Marks(int math, int science, int english) {
        this.math = math;
        this.science = science;
        this.english = english;
    }

    public int getTotalMarks() {
        return math + science + english;
    }

    public double getAverage() {
        return getTotalMarks() / 3.0;
    }

    public void showMarks() {
        System.out.println("Math: " + math);
        System.out.println("Science: " + science);
        System.out.println("English: " + english);
        System.out.println("Total Marks: " + getTotalMarks());
        System.out.println("Average Marks: " + getAverage());
    }
}
```

Ln 28, Col 1 | 693 characters | 160% | Windows (CRLF) | UTF-8

Student.java | Marks.java | ReportCard.java | SchoolSystem.java

File Edit View

```java
package student;

public class ReportCard {
    Student student;
    Marks marks;

    public ReportCard(Student student, Marks marks) {
        this.student = student;
        this.marks = marks;
    }

    public void generateReport() {
        System.out.println("===== REPORT CARD =====");
        student.showStudentInfo();
        marks.showMarks();
        System.out.println("=======================");
    }
}
```

Student.java | Marks.java | ReportCard.java | SchoolSystem.java

File Edit View

```java
import student.Student;
import student.Marks;
import student.ReportCard;

public class SchoolSystem {
    public static void main(String[] args) {
        // Creating a student
        Student s1 = new Student("Rahul", 101);

        // Assigning marks
        Marks m1 = new Marks(85, 90, 78);

        // Generating report card
        ReportCard report = new ReportCard(s1, m1);
        report.generateReport();
    }
}
```

```
C:\Java programs\Student>javac -d . Student.java

C:\Java programs\Student>javac -d . Marks.java

C:\Java programs\Student>javac -d . ReportCard.java

C:\Java programs\Student>javac SchoolSystem.java

C:\Java programs\Student>java SchoolSytem
Error: Could not find or load main class SchoolSytem
Caused by: java.lang.ClassNotFoundException: SchoolSytem

C:\Java programs\Student>java SchoolSystem
===== REPORT CARD =====
Student Name: Rahul
Roll Number: 101
Math: 85
Science: 90
English: 78
Total Marks: 253
Average Marks: 84.33333333333333
=======================

C:\Java programs\Student>
```

5. Online Shopping System

Problem Statement: Design a package shopping with classes like Product, Cart, and Order to handle product listings, adding to cart, and order processing.

```java
package shopping;

public class Product {
    String name;
    double price;
    int productId;

    public Product(int productId, String name, double price) {
        this.productId = productId;
        this.name = name;
        this.price = price;
    }

    public void showProduct() {
        System.out.println("Product ID: " + productId);
        System.out.println("Product Name: " + name);
        System.out.println("Price: $" + price);
    }
}
```

Product.java Cart.java Order.java ShoppingSystem.java

File Edit View

```java
package shopping;

public class Cart {
    Product[] products;
    int count; // To track the number of products in the cart

    public Cart(int size) { // Define cart size
        products = new Product[size];
        count = 0;
    }

    public void addProduct(Product product) {
        if (count < products.length) {
            products[count] = product;
            count++;
            System.out.println(product.name + " added to cart.");
        } else {
            System.out.println("Cart is full! Cannot add more products.");
        }
    }
```

```java
    public double getTotalPrice() {
        double total = 0;
        for (int i = 0; i < count; i++) {
            total += products[i].price;
        }
        return total;
    }

    public void showCart() {
        if (count == 0) {
            System.out.println("Cart is empty.");
            return;
        }
        System.out.println("Cart Items:");
        for (int i = 0; i < count; i++) {
            products[i].showProduct();
            System.out.println("-----------------");
        }
        System.out.println("Total Price: " + getTotalPrice());
    }
}
```

Ln 13, Col 39 | 1,134 characters | 160% | Windows (CRLF) | UTF-8

Product.java Cart.java Order.java ShoppingSystem.java

File Edit View

```java
package shopping;

public class Order {
    Cart cart;

    public Order(Cart cart) {
        this.cart = cart;
    }

    public void placeOrder() {
        if (cart.count == 0) {
            System.out.println("Cannot place order. Cart is empty!");
            return;
        }
        System.out.println("Order placed successfully!");
        cart.showCart();
        System.out.println("Thank you for shopping with us!");
    }
}
```

Product.java Cart.java Order.java ShoppingSystem.java

File Edit View

```java
import shopping.Product;
import shopping.Cart;
import shopping.Order;

public class ShoppingSystem {
    public static void main(String[] args) {
        Product p1 = new Product(101, "Laptop", 55000.0);
        Product p2 = new Product(102, "Smartphone", 20000.0);
        Product p3 = new Product(103, "Headphones", 2500.0);

        Cart cart = new Cart(5);
        cart.addProduct(p1);
        cart.addProduct(p2);
        cart.addProduct(p3);

        // Displaying cart details
        cart.showCart();

        // Placing an order
        Order order = new Order(cart);
        order.placeOrder();
    }
}
```

Command Prompt

```
C:\Java programs\Shopping>javac -d . Product.java

C:\Java programs\Shopping>javac -d . Cart.java

C:\Java programs\Shopping>javac -d . Order.java

C:\Java programs\Shopping>javac ShoppingSystem.java

C:\Java programs\Shopping>java ShoppingSystem
Laptop added to cart.
Smartphone added to cart.
Headphones added to cart.
Cart Items:
Product ID: 101
Product Name: Laptop
Price: $55000.0
-----------------
Product ID: 102
Product Name: Smartphone
Price: $20000.0
-----------------
Product ID: 103
Product Name: Headphones
Price: $2500.0
-----------------
Total Price: 77500.0
Order placed successfully!
Cart Items:
```

```
-----------------
Total Price: 77500.0
Order placed successfully!
Cart Items:
Product ID: 101
Product Name: Laptop
Price: $55000.0
-----------------
Product ID: 102
Product Name: Smartphone
Price: $20000.0
-----------------
Product ID: 103
Product Name: Headphones
Price: $2500.0
-----------------
Total Price: 77500.0
Thank you for shopping with us!

C:\Java programs\Shopping>
```

6. Vehicle Rental System

Problem Statement: Develop a package rental with classes such as Vehicle, RentalTransaction, and Customer to manage vehicle rentals, payments, and bookings.

```java
package rental;

public class Vehicle {
    String vehicleId;
    String model;
    double rentalPricePerDay;
    boolean isAvailable;

    public Vehicle(String vehicleId, String model, double rentalPricePerDay) {
        this.vehicleId = vehicleId;
        this.model = model;
        this.rentalPricePerDay = rentalPricePerDay;
        this.isAvailable = true; // Vehicle is available initially
    }

    public void showVehicleDetails() {
        System.out.println("Vehicle ID: " + vehicleId);
        System.out.println("Model: " + model);
        System.out.println("Rental Price Per Day: $" + rentalPricePerDay);
        System.out.println("Availability: " + (isAvailable ? "Available" : "Not Available"));
    }

    public boolean rentVehicle() {
        if (isAvailable) {
            isAvailable = false;
            return true;
        }
        return false;
    }

    public void returnVehicle() {
        isAvailable = true;
    }
}
```

```java
package rental;

public class Customer {
    String customerId;
    String name;
    String phoneNumber;

    public Customer(String customerId, String name, String phoneNumber) {
        this.customerId = customerId;
        this.name = name;
        this.phoneNumber = phoneNumber;
    }

    public void showCustomerDetails() {
        System.out.println("Customer ID: " + customerId);
        System.out.println("Name: " + name);
        System.out.println("Phone Number: " + phoneNumber);
    }
}
```

Vehicle.java | Customer.java | RentalTransaction.java | RentalSystem.java

File Edit View

```java
package rental;

public class RentalTransaction {
    Customer customer;
    Vehicle vehicle;
    int rentalDays;
    double totalAmount;

    public RentalTransaction(Customer customer, Vehicle vehicle, int rentalDays) {
        this.customer = customer;
        this.vehicle = vehicle;
        this.rentalDays = rentalDays;
        this.totalAmount = rentalDays * vehicle.rentalPricePerDay;
    }

    public void processRental() {
        if (vehicle.rentVehicle()) {
            System.out.println("\nRental Confirmed!");
            customer.showCustomerDetails();
            vehicle.showVehicleDetails();
            System.out.println("Rental Duration: " + rentalDays + " days");
            System.out.println("Total Amount: $" + totalAmount);
        } else {
            System.out.println("Sorry, the vehicle is not available for rent.");
        }
    }
```

Ln 33, Col 1 | 1,026 characters | 150% | Windows (CRLF) | UTF-8

```java
    public void returnVehicle() {
        vehicle.returnVehicle();
        System.out.println("\nVehicle returned successfully by " + customer.name);
    }
}
```

Ln 33, Col 1 | 1,026 characters | 150% | Windows (CRLF) | UTF-8

Vehicle.java | Customer.java | RentalTransaction.java | RentalSystem.java

File Edit View

```java
import rental.Vehicle;
import rental.Customer;
import rental.RentalTransaction;

public class RentalSystem {
    public static void main(String[] args) {
        // Creating vehicles
        Vehicle v1 = new Vehicle("V001", "Toyota Corolla", 50.0);
        Vehicle v2 = new Vehicle("V002", "Honda Civic", 60.0);

        // Creating customers
        Customer c1 = new Customer("C101", "Alice", "9876543210");
        Customer c2 = new Customer("C102", "Bob", "9123456780");

        // Processing rental transaction
        RentalTransaction transaction1 = new RentalTransaction(c1, v1, 3);
        transaction1.processRental();

        // Returning the vehicle
        transaction1.returnVehicle();

        // Another rental attempt
        RentalTransaction transaction2 = new RentalTransaction(c2, v1, 5);
        transaction2.processRental();
    }
}
```

Ln 3, Col 32 | 858 characters | 150% | Windows (CRLF) | UTF-8

```
C:\Java programs\Rental>javac -d . Vehicle.java

C:\Java programs\Rental>javac -d . Customer.java

C:\Java programs\Rental>javac -d . RentalTransaction.java

C:\Java programs\Rental>javac RentalSystem.java

C:\Java programs\Rental>java RentalSystem

Rental Confirmed!
Customer ID: C101
Name: Alice
Phone Number: 9876543210
Vehicle ID: V001
Model: Toyota Corolla
Rental Price Per Day: $50.0
Availability: Not Available
Rental Duration: 3 days
Total Amount: $150.0

Vehicle returned successfully by Alice
```

```
Rental Confirmed!
Customer ID: C102
Name: Bob
Phone Number: 9123456780
Vehicle ID: V001
Model: Toyota Corolla
Rental Price Per Day: $50.0
Availability: Not Available
Rental Duration: 5 days
Total Amount: $250.0

C:\Java programs\Rental>
```